Statistical Modeling - 24DS636 (2024-25)

Abhijith M S

2025-08-31

Table of contents

Chapter 1

Engineering_Graphics

This is a Quarto website.

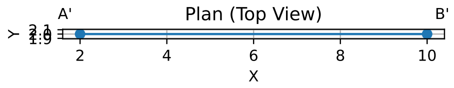
To learn more about Quarto websites visit https://quarto.org/docs/websites.

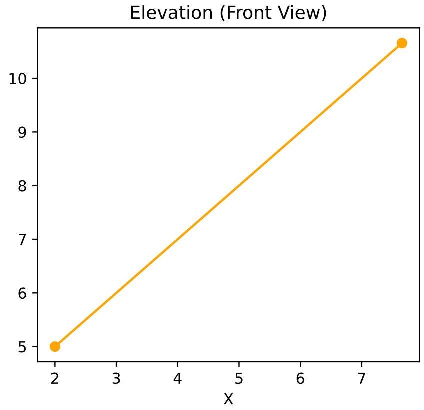
Chapter 2

Projection of Lines

```
import matplotlib.pyplot as plt
import numpy as np

# Line parameters
length = 8  # true length of the line
theta = 45  # inclination with VP in degrees

# Plan (top view): line parallel to XY (since parallel to HP)
x_plan = [2, 2 + length]
y_plan = [2, 2]

# Elevation (front view): inclined to XY by theta
x_elev = [2, 2 + length * np.cos(np.radians(theta))]
y_elev = [5, 5 + length * np.sin(np.radians(theta))]

fig, axs = plt.subplots(1, 2, figsize=(10, 4))

# Plan view
axs[0].plot(x_plan, y_plan, marker='o')
axs[0].set_title('Plan (Top View)')
axs[0].set_xlabel('X')
axs[0].set_ylabel('Y')
axs[0].set_aspect('equal')
axs[0].grid(True)
axs[0].annotate('A\'', (x_plan[0], y_plan[0]), textcoords=”offset points”, xytext=(-10,10), ha='center')
axs[0].annotate('B\'', (x_plan[1], y_plan[1]), textcoords=”offset points”, xytext=(10,10), ha='center')

# Elevation view
axs[1].plot(x_elev, y_elev, marker='o', color='orange')
axs[1].set_title('Elevation (Front View)')
axs[1].set_xlabel('X')
```

Text(0.5, 0, 'X')


Plan (Top View)
A'
B'
Y
2.1
2.0
1.9
2
4
6
8
10
X


Elevation (Front View)
10
9
8
7
6
5
2
3
4
5
6
7
X